Module 4: Padding and Stride

1. Why do we need padding and stride?

When a filter moves across an image, the output dimensions depend on the filter size and how far the filter moves.

Padding and stride help control the output size.

2. Padding

Padding means adding extra values around an image or feature map. Zero padding is a common method.

Original image:

1 2 3

4 5 6

7 8 9

Image with one-pixel zero padding:

0 0 0 0 0

0 1 2 3 0

0 4 5 6 0

0 7 8 9 0

0 0 0 0 0

Types of padding

Valid padding: No padding is added. The output generally becomes smaller.

Same padding: Padding is selected to preserve height and width when stride is 1.

Padding allows filter operations near the image boundaries and can help preserve spatial dimensions.

It does not guarantee that every pixel is used the same number of times.

3. Stride

Stride defines how many positions the filter moves at each step.

Stride 1: The filter moves one position at a time.

Stride 2: The filter moves two positions at a time.

Stride 1:

Filter moves one position at a time.

Stride 2:

Filter moves two positions at a time.

A larger stride generally creates a smaller feature map and reduces subsequent computation. However, it can also remove spatial detail from the representation.

Stride reduces spatial resolution, not directly the quality of the original image.

4. Output-size formula

For one spatial dimension, assuming dilation 1:

Output size = floor((N - F + 2P) / S) + 1

Where:

- N = input dimension
- F = filter dimension
- P = padding on each side
- S = stride

Height and width are calculated separately.

5. Example

Input image: 5 × 5

Filter: 3 × 3

Padding: 0

Stride: 1

Calculation:

Output size = (5 - 3 + 0) / 1 + 1

Output size = 3

Therefore, the output feature map is 3 × 3.

Now use padding = 1 and stride = 1:

Output size = (5 - 3 + 2) / 1 + 1

Output size = 5

Therefore, the output feature map is 5 × 5.

Conclusion

Padding controls the border treatment, while stride controls the distance the filter moves. Both affect the output dimensions of convolution.